

VII Comp. SC
19-5-14.

QP Code :MV-19980

(3 Hours)

[Total Marks : 100

- N. B. : (1) Question No. 1 is **compulsory**.
(2) Attempt any **four** questions out of remaining **six** questions.
(3) **Figures to the right** indicate **full marks**.

1. (a) Explain fuzzy extension principle with the help of an example. 6
(b) Model the following as a fuzzy set using suitable membership function - "numbers close to 10". 6
(c) Explain standard fuzzy membership functions. 8
2. Design a fuzzy controller to determine the wash time of domestic washing machine. Assume that the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptors for the output variable. Device a set of rules for control action and defuzzification. The design should be supported by figures wherever possible. Clearly indicate that if the clothes are soiled to a larger degree the wash time required will be more. 20
3. (a) What is learning? Compare different learning rules. 10
(b) Explain error back propagation training algorithm with the help of a flowchart. 10
4. (a) Implement the perceptron rule training using $f(\text{net}) = \text{sgn}(\text{net})$, $c=1$, and the following data specifying the initial weights $W1$, and the two training pairs. 10
 $W1 = [0, 1, 0]^t$
 $X1 = [2, 1, -1]^t$ $d1 = -1$;
 $X2 = [0, 1, -1]^t$ $d2 = 1$;
Repeat the training sequence until two correct responses in a row are achieved.
(b) Explain Hebbian Learning with the help of an example. 10
5. (a) Explain with examples linearly separable and non-linearly separable pattern classification. 10
(b) Explain the architecture of ANFIS with the help of a diagram. 10
6. (a) Explain with an example Genetic Algorithm. 10
(b) Explain RBF network and give the comparison between RBF and MLP. 10
7. Write a short note on **any two** of the following 20
(a) Derivative based Optimization
(b) Learning vector quantization
(c) Character recognition using neural network
(d) Kohonen's self organizing network.